# Oracle Cloud Infrastructure — Security Lists & Security Rules (Networking)

Documentação de referência oficial (consolidada) — Base de conhecimento para o Alura Agente

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## Contents



# Security Lists

The Networking service offers two virtual firewall features to control traffic at the packet level:

- **Security lists:** The original type of virtual firewall offered by the Networking service.
- **Network security groups (NSGs):** Another type of virtual firewall, recommended over security lists.

Both features use *security rules*.

## Highlights

- Security lists act as virtual firewalls for Compute instances and other kinds of resources. A security list consists of a set of ingress and egress security rules that apply to **all the VNICs in any subnet that the security list is associated with**. This means all VNICs in a particular subnet are subject to the same set of security lists.
- Security list rules function the same as network security group rules.
- Each VCN comes with a **default security list** that has several default rules for essential traffic. If you don’t specify a custom security list for a subnet, the default security list is automatically used with that subnet. You can add and remove rules from the default security list.
- Security lists have separate and different limits compared to network security groups.

## Overview of Security Lists

A security list acts as a virtual firewall for an instance, with ingress and egress rules that specify the types of traffic allowed in and out. Each security list is **enforced at the VNIC level**, but you **configure security lists at the subnet level** — all VNICs in a subnet are subject to the same set of security lists, whether communicating with another instance in the VCN or a host outside the VCN.

Each subnet can have several security lists associated with it (maximum 5), and each list can have many rules. A packet is allowed if **any rule in any of the associated lists** allows the traffic (or if the traffic is part of an existing tracked connection) — unless the lists happen to contain both stateful and stateless rules that cover the same traffic (in which case the stateless rule takes precedence).

Security lists are regional entities and can control both IPv4 and IPv6 traffic.

## Default Security List

Unlike other security lists, the default security list comes with an initial set of stateful rules. Oracle recommends changing these rules to only allow inbound traffic from authorized subnets relevant to the region.

- **Stateful ingress:** Allow TCP traffic on destination port **22 (SSH)** from authorized source IP addresses and any source port. This is why you can create a VCN, a public subnet, launch a Linux instance, and immediately SSH into it without writing any security rules yourself.
- **Important:** The default security list does **not** include a rule for RDP (port 3389). If using Windows images, you must add that rule manually.
- **Stateful ingress:** Allow ICMP traffic type 3 code 4 from authorized source IP addresses (enables Path MTU Discovery fragmentation messages).
- **Stateful ingress:** Allow ICMP traffic type 3 (all codes) from the VCN's CIDR block (lets instances receive connectivity error messages from other instances in the VCN).
- **Stateful egress:** Allow all traffic. Instances can start traffic of any kind to any destination; because of connection tracking, response traffic is automatically allowed regardless of ingress rules.

The default security list comes with **no stateless rules**, but you can add or remove any rule.

### Enabling Ping

The default security list does **not** include a rule to allow ping (ICMP echo) requests — you must add one explicitly if you want to ping an instance.

# Security Rules

Both security lists and network security groups (NSGs) use *security rules* to control packet-level traffic.

## Comparison of Security Lists and Network Security Groups

- **Security lists** define a set of rules that applies to **all VNICs in an entire subnet**. A subnet can be associated with a maximum of five security lists.
- **Network security groups (NSGs)** define a set of rules that applies to a **chosen group of VNICs** (e.g., all instances with the same security posture). A VNIC can be added to a maximum of five NSGs.

| Security tool | Applies to | To enable | Limitations |
|---|---|---|---|
| Security lists | All VNICs in a subnet using that security list | Associate the security list with the subnet | Max 5 security lists per subnet |
| Network security groups | Chosen VNICs in the same VCN | Add specific VNICs to the NSG | Max 5 NSGs per VNIC |

**Oracle recommends using NSGs instead of security lists**, because NSGs let you separate the VCN's subnet architecture from the application's security requirements, and Oracle prioritizes NSGs when implementing future enhancements. However, you can use both together — the set of rules that applies to a VNIC is the **union** of the security lists on its subnet and the NSGs it belongs to.

## Limits

### Security List Limits

| Resource | Scope | Limit |
|---|---|---|
| Security lists | VCN | 300 |
| Security lists | Subnet | 5 |
| Security rules | Security list | 200 ingress + 200 egress |

### Network Security Group Limits

| Resource | Scope | Limit |
|---|---|---|
| Network security groups | VCN | 1000 |
| Security rules | Network security group | 120 total (ingress + egress) |

## Parts of a Security Rule

A security rule **allows** a particular type of traffic in or out of a VNIC. Without security rules, no traffic is allowed in or out. Each rule specifies:

- **Direction:** Ingress (inbound) or egress (outbound).
- **Stateful or stateless:** Stateful uses connection tracking; stateless does not (see below).
- **Source type and source (ingress only):** CIDR block (e.g., `0.0.0.0/0` for all IPs), a Network Security Group (NSG rules only), or a Service (traffic from an Oracle service via a service gateway).

- **Destination type and destination (egress only):** Same options as source, applied to outbound traffic.
- **IP Protocol:** A single protocol, or “all.”
- **Source port / Destination port:** For TCP or UDP — all ports, a single port, or a range.
- **ICMP type and code:** For ICMP traffic — all types/codes, or a specific type with optional code.
- **Description:** Optional, NSG rules only (not supported for security list rules).

**Note:** Security rules are not enforced for traffic involving the `169.254.0.0/16` CIDR block (includes services like iSCSI and instance metadata).

## Stateful Compared to Stateless Rules

The default is **stateful**. Stateless is recommended for high-volume internet-facing websites (HTTP/HTTPS traffic).

### Stateful Rules

Marking a rule as stateful means Oracle uses **connection tracking** for traffic matching that rule. When an instance receives traffic matching a stateful ingress rule, the response is automatically tracked and allowed back to the originating host — **no corresponding egress rule is needed**. Likewise, for a stateful egress rule, the incoming response is automatically allowed without needing an ingress rule.

Example: a stateful ingress rule allowing HTTP traffic (TCP, all source IPs `0.0.0.0/0`, destination port 80) is sufficient on its own — you don’t need a separate egress rule for the response.

**Note:** Stateful rules store connection state in a tracking table on each Compute instance. Table size depends on the instance’s shape. If the table fills up, new connections experience packet loss. High-traffic subnets should consider stateless rules instead.

### Stateless Rules

Marking a rule as stateless means **no connection tracking** is used — response traffic is **not** automatically allowed. To allow the response for a stateless ingress rule, you must create a **corresponding stateless egress rule** (and vice versa).

Example: a stateless ingress rule allowing HTTP traffic on destination port 80 needs a matching stateless egress rule (destination `0.0.0.0/0`, any port, from source port 80) for the response to actually reach the client.

**Note:** If both a stateful and stateless rule match the same traffic in one direction, the **stateless rule takes precedence** and the connection is not tracked — you then need a corresponding rule in the other direction for the response to be allowed.

### Connection Tracking Details (Stateful Rules)

Oracle tracks: protocol, source/destination IP, and (for TCP/UDP) source/destination ports. For other protocols, only protocol and IP are tracked. Idle timeouts before a tracked connection is removed:

| Protocol | State | Idle timeout |
|---|---|---|
| TCP | Established | 1 day |
| TCP | Setup | 1 minute |
| TCP | Closure | 2 minutes |
| UDP | Established (both directions) | 3 minutes |
| UDP | Not established (one direction only) | 30 seconds |

| Protocol | State | Idle timeout |
|---|---|---|
| ICMP | N/A | 15 seconds |
| Other | N/A | 5 minutes |

## Enabling Path MTU Discovery for Stateless Rules

If using stateless rules for traffic to/from outside the VCN, add a rule allowing ingress ICMP type 3 code 4 from `0.0.0.0/0`. This enables Path MTU Discovery fragmentation messages and is critical for establishing connections — without it you can experience connectivity issues (hanging connections).

## Rules to Enable Ping

The default security list has no rule allowing ping. To ping an instance, add an extra stateful ingress rule allowing ICMP type 8 from the source network (use `0.0.0.0/0` to allow ping from the internet).

## Rules to Handle Fragmented UDP Packets

If a UDP packet is too large, it gets fragmented, and only the first fragment carries protocol/port information. If your security rules specify a particular port number, later fragments (which lack that info) get dropped. If you expect large UDP packets, set source and destination ports to **ALL** instead of a specific port.

## Troubleshooting Checklist — "Why Can't I Reach My Instance?"

When troubleshooting access to an instance, confirm **all** of the following are set correctly (any one of them can silently block traffic even if the others are correct):

1. **The rules in the network security groups** the instance's VNIC is in.
2. **The rules in the security lists** associated with the instance's subnet.
3. **The instance's own OS firewall rules** — Compute images running platform Linux distributions ship with OS-level firewalls (e.g., `iptables`/`firewalld`) *in addition to* the OCI-level security list/NSG. Opening a port in the Console is not enough if the OS firewall inside the VM still blocks it.

For instances running Oracle Autonomous Linux, Oracle Linux 7/8, or Oracle Linux Cloud Developer, use `firewalld` to manage the OS-level rules. Example — opening port 1521:

```
sudo firewall-cmd --zone=public --permanent --add-port=1521/tcp
sudo firewall-cmd --reload
```

**Best practices:**

- Get familiar with the default security list's rules before removing any of them — they exist to enable basic connectivity, and removing them without checking dependencies can break connectivity for existing resources.
- Prefer NSGs over security lists for new architectures, since Oracle recommends and prioritizes NSGs.
- Use the Networking service's VNIC metrics (ingress/egress packets dropped due to security rules) to diagnose whether traffic is actually being blocked at the security-rule layer.

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*Fim do documento consolidado. Fonte original: Oracle Cloud Infrastructure Documentation (docs.oracle.com), páginas "Security Lists" e "Security Rules".*